

R2.1 How much? The amount of chemical change

Checklist

What you should know

After studying this subtopic, you should be able to:

- Deduce chemical equations, including state symbols, when reactants and products are specified.
- Calculate reacting masses and/or volumes of reactants and products.
- Calculate the concentrations of reactants and products for reactions occurring in solution.
- Identify the limiting and excess reactants from given data.
- Distinguish between the theoretical yield and the experimental yield and suggest errors which may cause the experimental yield to be either higher or lower than the theoretical yield.
- Solve problems involving reacting quantities, limiting and excess reactants, theoretical, experimental and percentage yields.
- Calculate the atom economy from the stoichiometry of a reaction using the equation provided for you in the DP chemistry data booklet.
- Discuss the inverse relationship between atom economy and wastage in industrial processes.

Practical skills

Once you have completed this subtopic, go to **Practical 7: Synthesis of aspirin** (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/synthesis-of-aspirin-id-46723/>) in which you will create a complex chemical, calculate the yield and conduct purity tests.

